

NICHOLAS CALLAHAN

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OBJECTIVE

Recent graduate from Eastern Michigan University with a B.S. in Applied Computer Science and hands-on experience in software engineering, web automation, and artificial intelligence. Seeking an internship or junior-level position in software development, web engineering, or other areas of computer science to apply strong foundations in algorithms, data structures, and programming languages to real-world solutions.

EDUCATION

Eastern Michigan University 2022 – 2026

B.S. in Computer Science – Applied | Ypsilanti, MI

- Honors College | Dean's List
- Relevant Coursework: Algorithms & Data Structures, Introduction to Artificial Intelligence, Software Engineering, Computer Operating Systems, Programming Languages, Computational Discrete Structures, Computer Organization, Internet-Based Computing, Software Dev for Mobile Devices, Embedded AI, Computer Game Programming

TECHNICAL SKILLS

Languages: Python, Java, HTML/CSS, C/C#/C++, Exposure to JavaScript/TypeScript

Frameworks & Tools: Selenium WebDriver, Git/GitHub, Linux, REST APIs, Pytest

Environments: PyCharm, Eclipse, Visual Studio, BlueJ, Microsoft Office Suite, Google Workspace, Android Studio, Git Bash, VirtualBox, DOSBox, Arduino IDE, Postman

Concepts: Data Structures & Algorithms, Artificial Intelligence, OOP, Operating Systems, Software Engineering (Agile/Scrum), Discrete Mathematics, Computer Architecture, Web Development, Embedded Devices, Android Development, Unity Development

PROJECTS

Connect 4 AI – Honors Contract Project COSC 461, Winter 2026

Java | Intro to Artificial Intelligence – Eastern Michigan University

- Developed a Connect 4 game engine in Java featuring two AI opponents powered by Min-Max and Monte Carlo Tree Search algorithms; applied Object-Oriented Programming (OOP) principles and design patterns to structure game engine architecture for maintainability.
- Built a simulation framework to pit the algorithms against each other over hundreds of games, analyzing win-rate trends under constrained variables; implemented unit testing to validate AI algorithm performance under various constraints.
- Authored paper documenting algorithm performance, implementation challenges, and comparative analysis of search strategies

nn_dataflow Tools - Final Project COSC 480, Winter 2026

Python, Tangram Python Tool | Embedded AI - Eastern Michigan University

https://github.com/rpiza12/nn_dataflow/tree/master/nn_dataflow/visualization

- Extended an open-source neural network dataflow scheduling tool by implementing visualization scripts that parse JSON search results and generate per-layer energy, latency, and cost breakdowns using Matplotlib
- Defined a full MobileNetV2 architecture within the existing layer API, enabling dataflow scheduling on a modern mobile-optimized network not previously supported by the tool
- Validated the new network definition against the original MobileNetV2 paper's published layer dimensions and verified correctness through the project's built-in test suite

Bandcamp Auto-Checkout Script Personal Project

Python, Selenium WebDriver

- Built a web automation tool to monitor Bandcamp product pages and execute rapid auto-checkout for limited-release vinyl and merchandise drops
- Implemented configurable wait logic, DOM element targeting, and automated form-fill for seamless end-to-end purchasing
- Designed retry mechanisms and error handling to manage dynamic page loads and rate-limiting

EXPERIENCE

Lowe's – Indoor Garden Sales Associate 2023 – Present

Howell, MI

- Assist customers in selecting power equipment, tools, and plants; use store computer and phone systems to locate products for customers, and assist with checkout
- Operate specialized power equipment for stocking and merchandising while maintaining safety standards

Levi Strauss & Co. – Stylist 2022 – 2023

Howell, MI

- Managed point-of-sale transactions and delivered a smooth customer checkout experience
- Conducted inventory management, including restocking and fulfilling online orders

Clothes Mentor – Online Sales Associate 2022

Novi, MI

- Managed the e-commerce storefront using Shopify, posting products and promoting via Instagram and Facebook
- Handled order fulfillment and shipping logistics; supported overall online store operations

HONORS & INVOLVEMENT

Eastern Michigan University Honors College 2022 – 2026

- Maintaining academic distinction while completing enriched coursework; earned Dean's List recognition

Community Service – Livingston County Recycle Center 2026

- Regularly volunteered with the organization as a part of the Honors College civic engagement

References

Available upon request.